IET Image Processing

Research Article

Ring oscillator as confusion – diffusion agent: a complete TRNG drove image security

ISSN 1751-9659
Received on 8th February 2019
Revised 8th April 2020
Accepted on 12th May 2020
E-First on 5th October 2020
doi: 10.1049/iet-ipr.2019.0168
www.ietdl.org

Rethinam Sivaraman[1], Sundararaman Rajagopalan[1] ✉, John Bosco Balaguru Rayappan[1], Rengarajan Amirtharajan[1]

[1]*Department of ECE, School of EEE, SASTRA Deemed University, Thanjavur – 613401, India*
✉ *E-mail: raman@ece.sastra.edu*

Abstract: The utility of true random number generators (TRNGs) is not only restricted to session key generation, nonce generation, OTP generation etc. in cryptography. In the proposed work, two ring oscillator (RO) based TRNG structures adopting identical and non-identical ring of inverters have alone been employed for confusion (scrambling) and diffusion (intensity variation) processes for encrypting the greyscale and RGB images. Cyclone IVE EP4CE115F29C7 FPGA was utilised to generate a couple of random synthetic images using the two RO architectures which took a maximum of 520 combinational units and 543 logic registers. The suggested scheme of image encryption was tested on 100 test greyscale images of size 256 × 256. This non-chaos influenced image ciphering has resulted in an approximate average entropy of 7.99 and near-zero correlation figures for the greyscale & RGB cipher images. The attack resistance capability was checked by performing various occlusion and noise attacks on encrypted images.

1 Introduction

Image encryption plays an important role in safeguarding various image modalities [1]. Chaotic maps have occupied a significant place in encryption for a single round or multi-round confusion and diffusion processes [2–5]. Such maps namely logistic, tent, sine, combine chaotic etc. are being used extensively in the recent research works [6–12]. However, the non-uniform chaotic range provided by the maps needs critical attention. This selective chaotic region restricts the random range width even though the sensitivity to initial conditions forms the basis for time series generation through the chaotic maps or attractors. Chaos-based ciphering techniques are becoming vulnerable to attacks [13].

Recently, cryptanalysis has been performed by Norouzi and Mirzakuchaki [14] on chaos-based greyscale image encryption wherein the algorithm was compromised through the chosen plaintext attack. Dou *et al.* also performed a cryptanalysis on logistic maps blend DNA-based combined image encryption scheme. Their work was capable of cracking through a novel chosen-plaintext attack [15].

Random image or Synthetic image generated through the hardware seems to be an option for chaotic maps. Any strong alternative must be capable of providing the impact of confusion and diffusion yielded by the chaos influenced maps. Random bits acquired through unique features of hardware can resist the attacks due to the difficulties in arriving a particular mathematical model. In a DNA – Chaotic image encryption work proposed by Zhang *et al.* [16], a key image was generated using randint (⟨size⟩) command of Matlab 7.1. However, the correlation of this synthetic image was inferred to be low. Noticeably, true random number generators (TRNGs) have been incorporated in earlier works for the diffusion of images [17].

Rajagopalan *et al.* [18] suggested an ring oscillator (RO) based synthetic image captured using IC 7404 inverter rings. This near-uniform histogram yielding random image had an entropy of 7.954 close to 8. In another work, Sundararaman *et al.* [19] utilised a synthetic image generated through PLLs driven by beat frequency detection. They implemented this TRNG on Cyclone II FPGA which took 480 LEs for the formation of a synthetic image with an entropy of 7.99.

A five-layer DICOM medical image encryption proposed by Ravichandran *et al.* [20] incorporated a PLL-based synthetic image which consumed 98 hardware units for a generation. In all these works, chaotic maps/attractors played a significant role along with TRNG-based random images. Noticeably, the proposed method has utilised only the FPGA-based RO TRNG structures for both the confusion and diffusion operations without any chaotic schemes. The absence of floating-point units which will be required for hardware-based chaotic maps reduces the hardware consumption. Moreover, the entropy source adopted in this work does not need an initial seed or condition to trigger the generation of a random synthetic image.

2 Methodology

2.1 Synthetic image generation

In this image encryption scheme, clock jitters from ROs were utilised as a source of randomness to create the synthetic images which act as primary keys. The two different RO-based TRNG structures (Identical ROs and Non-identical ROs) adopted are shown in Figs. 1*a* and *b*, respectively. ROs are the promising agents for producing jittery clocks from which the jitters can be extracted using a harvesting mechanism to generate true random bits.

Fig. 1*a* illustrates the identical RO TRNG where it has 8 rings of 24 inverters. Jittery clock from each ring has been XORed to acquire the sequence of true random bits. Fig. 1*b* depicts the non-identical RO TRNG in which 64 inverters were used to form the TRNG structure and the bits from each ring were XORed to produce true random sequence [21]. Further, the generated true random bits from TRNGs need to be packed as 65,536 numbers of 8-bit width to form 256 × 256 synthetic images for the proposed application. Both the TRNGs have been designed using HDL and implemented on Altera Cyclone IV EP4CE115F29C7 FPGA using Altera Quartus 13.0 Electronic Design Automation tool.

Hardware analyses shown in Table 1 present the performance evaluation of TRNGs on Cyclone IV FPGA. Synthetic image of identical RO TRNG (Fig. 2*a*) achieves an entropy of 7.9368 with correlation figures of 0.0038, −0.0017 and 0.0106 whereas non-identical RO TRNG-based synthetic image (Fig. 2*b*) yields an entropy of 7.9701 and correlation metrics of −0.00047, −0.00081 and 0.0051. As per the timing analysis performed using the Logic

© The Institution of Engineering and Technology 2020

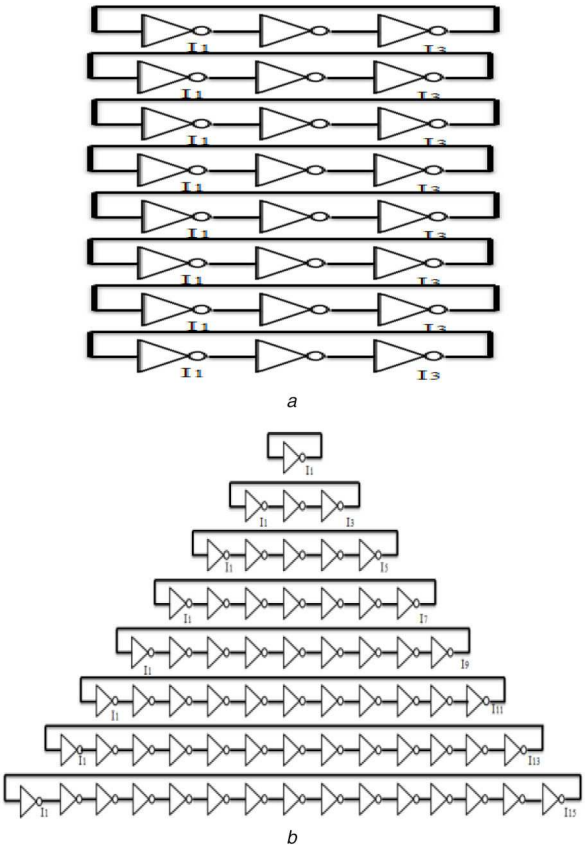


Fig. 1 *Entropy source*
(a) Identical RO TRNG, ***(b)*** Non-identical RO TRNG

Table 1 Hardware analysis: RO TRNGs

Target device		Cyclone IV E EP4CE115F29C7 FPGA	
TRNG structures		**identical Rings TRNG**	**non-identical Rings TRNG**
logic elements	combinational functions	518/1,14,480	520/1,14,480
	logic registers	543/1,14,480	543/1,14,480
	memory bits	83,88,608	83,88,608
power dissipation, mW	static power	98.68	98.68
	dynamic power	16.58	16.61
	IO power	32.31	32.31

Bold font is used for representing the headings.

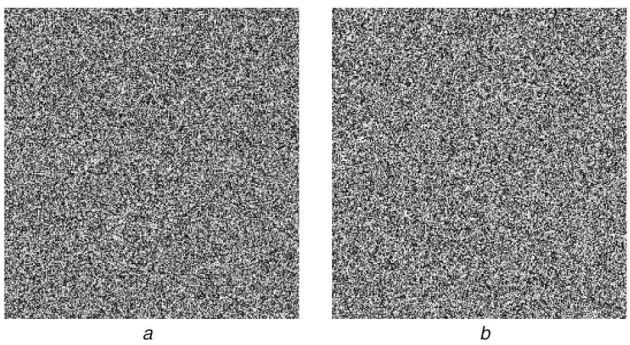


Fig. 2 *Synthetic image*
(a) Identical RO, ***(b)*** Non-identical RO

analyser, 42.6 ms was required to generate the synthetic image at a throughput of 12.307 Mbps.

2.2 Encryption procedure

In general, image encryption should have two unique operations, namely pixel scrambling and diffusion. Scrambling process breaks the correlation between pixels wherein the diffusion performs intensity modification. The proposed encryption of images can be performed through the following steps for greyscale as well as RGB images (see Fig. 3 and 4). Encryption process can be explained with a 4 × 4 matrix as follows:

Let the input 4×4 image matrix $\boldsymbol{I}(i,j)$, generated Identical RO synthetic 4×4 image matrix $\S_1\,(i,j)$ and generated Non-identical RO synthetic 4×4 image matrix $\S_2\,(i,j)$ be (see (1)) . Sorting the values of $\S_2\,(i,j)$ in ascending order, we get

$$\S_1(\text{sorted}) = \begin{bmatrix} 11 & 53 & 167 & 202 \\ 16 & 95 & 181 & 206 \\ 43 & 134 & 181 & 240 \\ 48 & 157 & 195 & 253 \end{bmatrix}$$

The location indices of sorted pixels of the matrix $\S_{1(\text{sorted})}$ in $\S_1(i,j)$ are given by

$$\S_1(\text{sorted_loc}) = \begin{bmatrix} 10 & 13 & 8 & 2 \\ 15 & 16 & 6 & 5 \\ 4 & 14 & 9 & 11 \\ 3 & 12 & 1 & 7 \end{bmatrix}$$

© The Institution of Engineering and Technology 2020

Scrambling the pixel positions of input image $I(i,j)$ using $\S_{1(\text{sorted_loc})}$, we get

$$\boldsymbol{I}\text{confused} = \begin{bmatrix} 155 & 155 & 158 & 162 \\ 158 & 157 & 155 & 153 \\ 159 & 153 & 156 & 155 \\ 164 & 158 & 162 & 157 \end{bmatrix}$$

Now, performing diffusion by bitwise XORing of $\boldsymbol{I}_{\text{confused}}$ (i,j) and $\S_2$ (i,j) to get final encrypted matrix (see equation below).

3 Results and discussion

The proposed image encryption scheme was implemented using LabVIEW 2013 tool with FPGA generated synthetic images as primary keys. The significance of this work is that the TRNG assisted synthetic images cannot be reproduced, which also possesses high randomness and keyspace to yield good encryption. The algorithm was tested on 100 greyscale images and 5 RGB images. Figs. 5*a–d* and 6*a–d* show four test greyscale images among the pool of images considered and their corresponding encrypted images. Figs. 7*a–d* and 8*a–d* present the four test colour images and their corresponding encrypted images. The sternness of the proposed encryption technique was evaluated through the statistical and chosen-plaintext attack analyses.

3.1 Statistical attack analyses

3.1.1 Entropy analysis: Entropy is defined as a measure of uncertainty which represents the degree of randomness in the encrypted image. For a good encrypted greyscale image, entropy must be close to 8. This work achieves an average entropy of 7.99654 close to the ideal value. Entropy can be calculated as [12]

$$E = \sum_{i=0}^{N-1} P(L_i)\log_2 P(Li) \tag{2}$$

where N is the number of intensity levels and $P(L_i)$ is the probability of each greyscale intensities in the whole image. Fig. 9 portrays the computed entropy of 100 greyscale images and an average value of 7.993518, which is close to the required entropy value 8, was inferred through this analysis.

3.1.2 Correlation analysis: Correlation between the adjacent pixels of an encrypted image must be close to zero. In this work, the synthetic image has been employed for the confusion which has resulted in near-zero correlation. The correlation coefficients have been computed in three different directions namely horizontal, vertical and diagonal. For the computation of correlation among the adjacent pixels, parameters namely mean, variance and

Step 1: Generate two synthetic images ($\S_1$, $\S_2$) of sizes 256 × 256 as explained in Section 2.1.
Step 2: Choose the original greyscale image ($\ddot{\boldsymbol{I}}$) of size 256 × 256.
Step 3: Sort the pixels in $\S_1$ and store the indices in the 1D array ($\Delta\S_1$) in a row matrix 1 × 65,536
Step 4: Restructure the 2D image matrix $\ddot{\boldsymbol{I}}$ to a 1D matrix and store the pixel values in an array ($\Delta\ddot{\boldsymbol{I}}$) of size 1 × 65,536.
Step 5: Perform the following scrambling operation to generate confused image $\Delta\hat{\boldsymbol{s}}$: $\Delta\hat{\boldsymbol{s}}\ \boldsymbol{i}) = \Delta\ddot{\boldsymbol{I}}\ (\Delta\S_1(j))$.
Step 6: Resize the $\Delta\hat{\boldsymbol{s}}$ of size to 1 × 65,536 to a scrambled image $\hat{\boldsymbol{s}}\ddot{\boldsymbol{I}}$ of size 256 × 256.
Step 7: Perform XOR operation between scrambled image $\hat{\boldsymbol{s}}\ddot{\boldsymbol{I}}$ and synthetic image $\S_2$: $\hat{\boldsymbol{E}}\ddot{\boldsymbol{I}} \rightarrow \hat{\boldsymbol{s}}\ddot{\boldsymbol{I}}$ XOR $\S_2$, where $\hat{\boldsymbol{E}}\ddot{\boldsymbol{I}}$ is the final encrypted image.
Step 8: Decryption is the reverse process of encryption to get back the original image.

Fig. 3 *Greyscale image encryption*

Step 1: Generate two synthetic images ($\S_1$, $\S_2$) of sizes 256 × 256 as explained in Section 2.1.
Step 2: Divide the colour image $\boldsymbol{RGB(x,y)}$ of size 256 × 256 into Red ($R(x,y)$), Green ($G(x,y)$) and Blue ($B(x,y)$) planes.
Step 3: Choose the Red plane $\boldsymbol{R(x,y)}$ of size 256 × 256.
Step 4: Sort the pixels in Synthetic image $\S_1$ and store the indices in 1D array ($\Delta\S_1$) as a row vector 1 × 65,536.
Step 5: Restructure the 2D image matrix $\boldsymbol{R(x,y)}$ as row vector and store the pixel values in an array ($\Delta\ddot{\boldsymbol{I}}$) of size 1 × 65,536.
Step 6: Perform the following scrambling operation to generate confused image $\Delta\hat{\boldsymbol{s}}$:
$\Delta\hat{\boldsymbol{s}}\ (\boldsymbol{i}) = \Delta\boldsymbol{R}(\boldsymbol{x},\boldsymbol{y})\ (\Delta\S_1(\boldsymbol{j}))$.
Step 7: Resize the row vector $\Delta\hat{\boldsymbol{s}}$ of size 1 × 65,536 into a scrambled image $\hat{\boldsymbol{s}}\boldsymbol{R}(\boldsymbol{x},\boldsymbol{y})$ of size 256 × 256.
Step 8: Perform XOR operation between scrambled image $\hat{\boldsymbol{s}}\boldsymbol{R}(\boldsymbol{x},\boldsymbol{y})$ and Synthetic image $\S_2$:
$\hat{\boldsymbol{E}}\boldsymbol{R}(\boldsymbol{x},\boldsymbol{y}) \rightarrow \hat{\boldsymbol{s}}\boldsymbol{R}(\boldsymbol{x},\boldsymbol{y})$ XOR $\S_2$, where $\hat{\boldsymbol{E}}\boldsymbol{R}(\boldsymbol{x},\boldsymbol{y})$ is the encrypted image of Red plane.
Step 9: Continue Steps 3 to 8 for $G(x,y)$ and $B(x,y)$ to the generate encrypted images of Green ($\hat{E}G(x,y)$) and Blue ($\hat{E}B(x,y)$) planes.
Step 10: Combine $\hat{E}R(x,y)$, $\hat{E}G(x,y)$ and $\hat{E}B(x,y)$ to get the encrypted RGB image.
Step 11: Decryption is the reverse process of encryption to get back the original image.

Fig. 4 *RGB image encryption*

$$\boldsymbol{I}(i,j) = \begin{bmatrix} 162 & 162 & 164 & 159 \\ 153 & 155 & 157 & 158 \\ 156 & 155 & 155 & 158 \\ 155 & 153 & 158 & 157 \end{bmatrix}, \S_1(i,j) = \begin{bmatrix} 195 & 202 & 48 & 43 \\ 206 & 181 & 253 & 167 \\ 181 & 11 & 240 & 157 \\ 53 & 134 & 16 & 95 \end{bmatrix} \&$$
$$\S_2(i,j) = \begin{bmatrix} 96 & 200 & 30 & 28 \\ 56 & 228 & 131 & 18 \\ 92 & 227 & 114 & 106 \\ 30 & 243 & 205 & 222 \end{bmatrix} \tag{1}$$

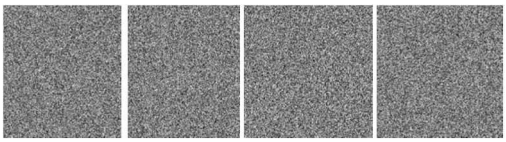
$$\boldsymbol{I}\text{encrypted} = \boldsymbol{I}\text{confused}\,(i,j) \oplus \S_2(i,j)$$
$$\boldsymbol{I}\text{encrypted} = \begin{bmatrix} 155 & 155 & 158 & 162 \\ 158 & 157 & 155 & 153 \\ 159 & 153 & 156 & 155 \\ 164 & 158 & 162 & 157 \end{bmatrix} \oplus \begin{bmatrix} 96 & 200 & 30 & 28 \\ 56 & 228 & 131 & 18 \\ 92 & 227 & 114 & 106 \\ 30 & 243 & 205 & 222 \end{bmatrix}$$
$$\boldsymbol{I}\text{encrypted} = \begin{bmatrix} 251 & 83 & 128 & 190 \\ 166 & 121 & 24 & 139 \\ 195 & 122 & 238 & 241 \\ 186 & 109 & 111 & 67 \end{bmatrix}$$

© The Institution of Engineering and Technology 2020

covariance are estimated for the chosen pixels of the image considered. The computation of correlation is expressed through (3)–(6) [13]. Tables 2 and 3 present the entropy and correlation results of greyscale and RGB images, respectively. Table 4 compares the entropy and correlation coefficients of original and encrypted images with respect to the earlier works. Also, Fig. 10 expresses the correlation analyses of 100 encrypted images. It can be inferred that the proposed algorithm achieves an average correlation value close to zero (see Figs. 11 and 12)

a b c d

Fig. 5 *Original greyscale images*
(a) Boat, ***(b)*** Cameraman, ***(c)*** Hydrangeas, ***(d)*** Peppers



a b c d

Fig. 6 *Encrypted greyscale images*
(a) Boat, ***(b)*** Cameraman, ***(c)*** Hydrangeas, ***(d)*** Peppers

$$E(x) = \frac{1}{N}\sum_{i=1}^{N} xi \tag{3}$$

$$D(x) = \frac{1}{N}\sum_{i=1}^{N} (xi - E(x))^2 \tag{4}$$

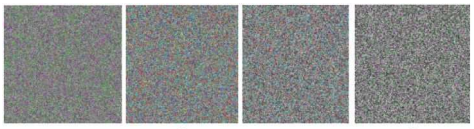
$$\mathrm{Cov}(x, y) = \frac{1}{N}\sum_{i=1}^{N} (xi - E(x))(y_i - E(y)) \tag{5}$$

$$\gamma_{xy} = \frac{\mathrm{Cov}(x, y)}{\sqrt{D(x)}\sqrt{D(y)}} \tag{6}$$

where $E(x)$ and $D(x)$ are the mean and variance of the selected number of pixels, respectively. $E(y)$ and $D(y)$ are the mean and variance of the adjacent pixels. Cov (x, y) is the co-variance computed with the variance values of x and y. Using the covariance and individual variances of x and y, the correlation (γ_{xy}) in a

a b c d

Fig. 7 *Original colour images*
(a) Airplane, ***(b)*** Lena, ***(c)*** Peppers, ***(d)*** Splash



a b c d

Fig. 8 *Encrypted colour images*
(a) Airplane, ***(b)*** Lena, ***(c)*** Peppers, ***(d)*** Splash

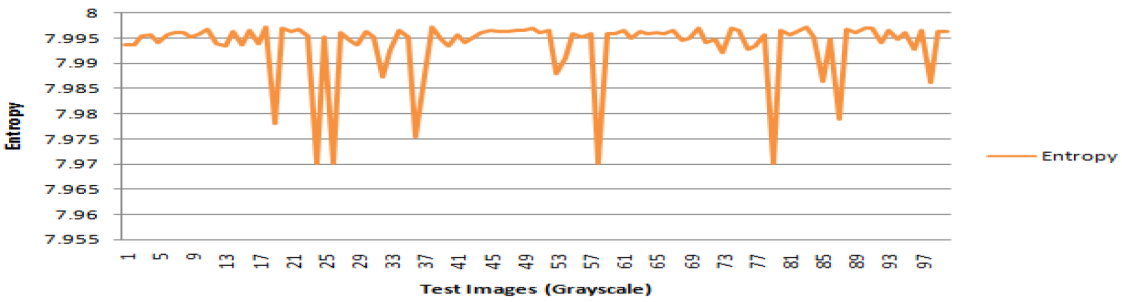


Fig. 9 *Entropy analysis for 100 greyscale test images*

Table 2 Statistical analysis – Greyscale images

Images		Entropy	*H*	*V*	*D*
Boat	original	7.2644	0.8460	0.8972	0.7984
	encrypted	7.9958	0.0073	0.0032	−0.0048
Cameraman	original	7.1289	0.9303	0.9590	0.9048
	encrypted	7.9962	0.0013	0.0013	0.0006
Hydrangeas	original	7.2123	0.9596	0.9660	0.9476
	encrypted	7.9966	0.0070	0.0062	−0.0036
Peppers	original	7.5768	0.9756	0.9760	0.9538
	encrypted	7.9969	0.0030	−0.0039	−0.0055
Lena	original	7.4582	0.9234	0.9724	0.9055
	encrypted	7.9972	0.0020	−0.0068	0.0036

E – Entropy, *H* – Horizontal, *V* – Vertical and *D* – Diagonal correlation coefficients.

© The Institution of Engineering and Technology 2020

Table 3 Statistical analysis – RGB images

Images		Entropy	*H*	*V*	*D*
Airplane	original (R)	6.7780	0.9055	0.8869	0.8252
	original (G)	6.8795	0.8923	0.8961	0.8309
	original (B)	6.2680	0.9088	0.8574	0.8215
	encrypted (R)	7.9927	−0.0033	−0.0006	−0.0010
	encrypted (G)	7.9914	0.0038	0.0020	−0.0001
	encrypted (B)	7.9931	−0.0016	0.0015	−0.0057
Lena	original (R)	7.2796	0.93338	0.9641	0.9074
	original (G)	7.6315	0.9049	0.9467	0.8802
	original (B)	6.9891	0.8597	0.9061	0.8383
	encrypted (R)	7.9959	−0.0137	0.0038	0.0036
	encrypted (G)	7.9961	−0.0049	−0.0008	−0.0012
	encrypted (B)	7.9959	0.0038	0.0017	0.0008
Peppers	original (R)	7.3319	0.9424	0.9464	0.9094
	original (G)	7.5242	0.9559	0.9617	0.9285
	original (B)	7.0793	0.9319	0.9406	0.8981
	encrypted (R)	7.9954	0.0019	0.0043	0.0029
	encrypted (G)	7.9944	0.0010	0.0052	−0.0018
	encrypted (B)	7.9939	0.0001	−0.0006	−0.0064
Splash	original (R)	6.9786	0.9824	0.9931	0.9775
	original (G)	6.8963	0.9550	0.9681	0.9350
	original (B)	6.1340	0.9572	0.9525	0.9224
	encrypted (R)	7.9941	0.0033	0.0026	0.0017
	encrypted (G)	7.9883	0.0018	−0.0034	−0.0003
	encrypted (B)	7.9944	0.0028	0.0032	0.0017
Tree	original (R)	7.2104	0.9590	0.9361	0.9159
	original (G)	7.4136	0.9687	0.9457	0.9318
	original (B)	6.9207	0.9612	0.9406	0.9265
	encrypted (R)	7.9955	−0.0038	−0.0044	0.0002
	encrypted (G)	7.9964	−0.0012	0.0060	−0.0040
	encrypted (B)	7.9952	−0.0083	0.0007	−0.0046

Table 4 Statistical analysis comparison – Greyscale images

Image		Entropy	*H*	*V*	*D*
Lena	**encrypted**	**7.9972**	**0.0020**	**−0.0068**	**0.0036**
[6]	encrypted	7.9912	0.0001	0.0089	0.0091
[11]	encrypted	7.9988	−0.0083	0.0040	−0.0463
[12]	encrypted	7.9030	0.0018	0.0011	0.0012
[9]	encrypted	7.9963	0.0094	0.0048	0.0112
[7]	encrypted	7.9971	0.0051	0.0038	0.0092
[1]	encrypted	7.9973	0.0010	0.0017	0.125
[3]	encrypted	7.9979	0.0010	0.0042	0.0063
[4]	encrypted	7.9969	0.0079	−0.0203	0.0049
[8]	encrypted	7.9971	0.0097	0.0178	0.0112
[5]	encrypted	7.9973	−0.0226	0.0041	0.0368
[2]	encrypted	7.9974	−0.0230	0.0019	−00034

Bold values indicate the obtained result of the proposed work.

specific direction can be computed. For this analysis, entire pixels of the images were considered. From the results, it can be inferred that the near-zero correlation has been obtained for encrypted images.

3.1.3 Histogram analysis: The histogram is the graphical representation of the probability distribution of pixel intensities in an image. For an encrypted image, the histogram plot indicates the degree of uniform distribution of intensity values. Figs. 13*a* and *b* show the histogram analyses of original and encrypted images of a test greyscale image and Figs. 14*a–c* and 15*a–c* present the histogram findings of an RGB image.

Figs. 11*a–c* portray the horizontal, vertical and diagonal correlation images of greyscale Pepper image, respectively, wherein Figs. 12*a–c* display their corresponding encrypted versions.

3.2 Keyspace analysis

It is a primary analysis to evaluate the key strength in terms of a number of possible random keys to resist the brute force attack. The keyspace of this proposed algorithm for the greyscale image has been calculated as follows:

Keyspace for confusion: $2^{8\times65536}$

Keyspace for diffusion: $2^{8\times65536}$

© The Institution of Engineering and Technology 2020

For a 256×256 greyscale image, the total keyspace has been calculated as $2^{(8\times65536+1)}$ which is greater than 2^{128} and also other earlier works [5–9, 11, 12] to resist the brute force attack.

3.3 Differential attack analysis

Number of Pixel Change Ratio (NPCR) and Unified Average Change in Intensity (UACI) are considered as two metrics to evaluate the sensitivity of any image encryption algorithm through differential attack analysis [16]. This analysis is operated on two cipher images ($C1$, $C2$) where $C1$ is the cipher image for the original image and $C2$ is the cipher image for the modified original image with one pixel change. The algorithm should produce different results for even a small change in the original image. Table 5 depicts the outcomes of differential analysis which satisfies the requirement of image encryption verified through the critical

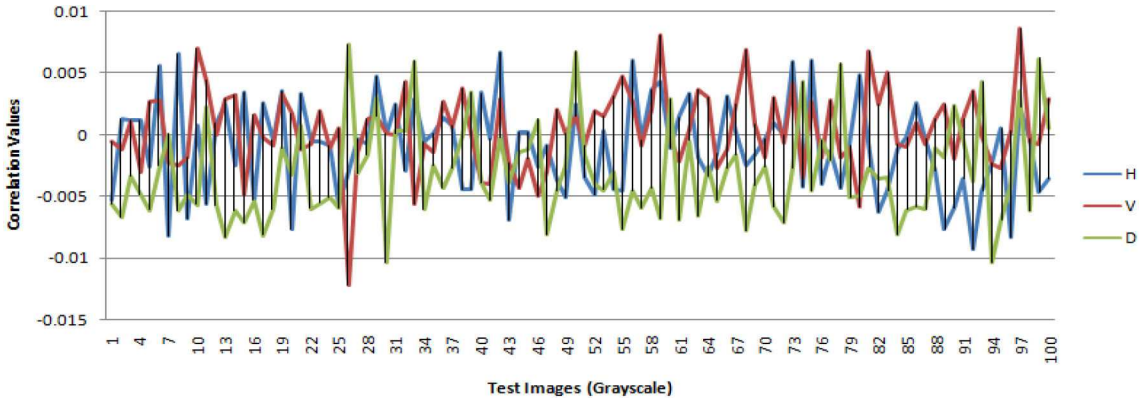


Fig. 10 *Correlation analysis for 100 greyscale test images*

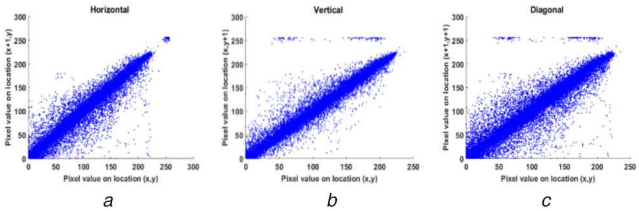


Fig. 11 *Correlation analysis: original greyscale Pepper image*
(a) Horizontal, ***(b)*** Vertical, ***(c)*** Diagonal

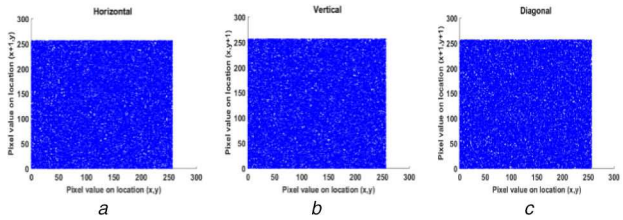


Fig. 12 *Correlation analysis: encrypted greyscale Pepper image*
(a) Horizontal, ***(b)*** Vertical, ***(c)*** Diagonal

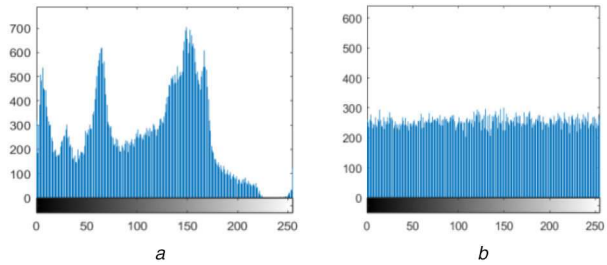


Fig. 13 *Histogram analysis on greyscale image – Peppers*
(a) Original, ***(b)*** Encrypted

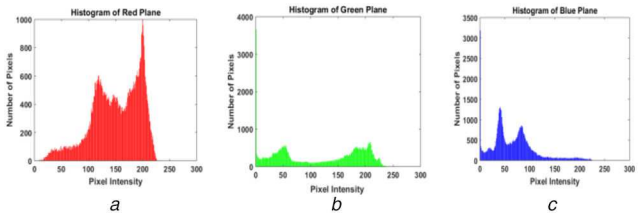


Fig. 14 *Histogram analysis of original RGB pepper image*
(a) Red plane, ***(b)*** Green plane, ***(c)*** Blue plane

© The Institution of Engineering and Technology 2020

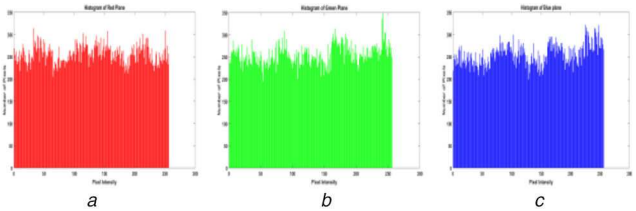


Fig. 15 *Histogram analysis on RGB image – Peppers*
(a), ***(b)***, ***(c)*** Encrypted

Table 5 Differential analysis

Images	NPCR	UACI
Boat	99.5758	32.2809
Cameraman	99.6094	28.7078
Hydrangeas	99.6460	31.3390
Peppers	99.5773	37.8632
Lena	99.6231	30.9135
[7]	99.59	33.48
[8]	98.9182	32.7865
[9]	99.6228	33.7041
[11]	99.6087	33.4584
[12]	99.6166	33.4365

Table 6 Encryption quality analysis – Greyscale images

Images	MD	ID	DH
Boat	5.4113×10^4	33,185	0.8207
Cameraman	61,193	30,902	0.9232
Hydrangeas	55,653	15,834	0.8535
Lena	46,254	21,261	0.7049
Peppers	3.7651×10^4	27,168	0.5764

values [16]. From Table 5, the proposed method has yielded acceptable results which are compared with other works.

3.4 Encryption quality analysis

To verify the effective encryption between the original and cipher images, three standard quality metrics have been considered. The degree of deviation of cipher image from its original image has been estimated through maximum deviation, irregular deviation and deviation from uniform histogram analyses. They can be expressed as

Maximum deviation [18]:

$$MD = \frac{D1 + DN - 1}{2} + \sum_{i=1}^{N-2} Di \tag{7}$$

Irregular deviation [18]:

$$ID = \sum_{i=0}^{N-1} HDi \tag{8}$$

Deviation from uniform Histogram [18]:

$$DH = \frac{\sum_{Ci=0}^{255} |Hci - Hc|}{M \times N} \tag{9}$$

From Tables 6 and 7, it can be observed that the deviation between original and encrypted images was high which confirms the efficiency of the proposed algorithm to produce a high degree of variation in the intensity levels. Further, this algorithm was tested on 100 greyscale images and the results are shown in Figs. 16*a* and *b*, which indicate maximum variations.

Table 7 Encryption quality analysis – Colour images

Images		MD	ID	DH
Airplane	red	68,589	46,019	1.0349
	green	6.7243×10^4	45,327	1.0510
	blue	7.9205×10^4	4919	1.2042
Lena	red	5.0069×10^4	45,906	0.7734
	green	36,473	25,140	0.5492
	blue	65,826	2750	0.9882
Peppers	red	52,294	39,102	0.8169
	green	3.7113×10^4	29,411	0.5934
	blue	59,754	16,950	0.9148
Splash	red	59,076	45,225	0.8987
	green	6.4346×10^4	17,634	1.0265
	blue	81,426	20,937	1.2462
Tree	red	47,634	34,116	0.7274
	green	4.6113×10^4	32,247	0.7087
	blue	63,673	37,040	0.9582

3.5 Error metric analyses

Mean square error (MSE) and Peak-signal-to-noise ratio (PSNR) are the two significant measures used for differentiating the original and cipher images [22]. Lower PSNR indicates the more degree of variation between original and encrypted images.

Figs. 17 and 18*a* and *b* illustrate the error metric analyses where the MSE and PSNR values are in the expected range ensuring the quality of encryption. In addition, Figs. 19*a* and *b* present the error metric analyses for 100 greyscale images which yielded an average MSE of 10914.3 and PSNR of 17.896 dB thereby indicating the effectiveness of the proposed encryption scheme.

© The Institution of Engineering and Technology 2020

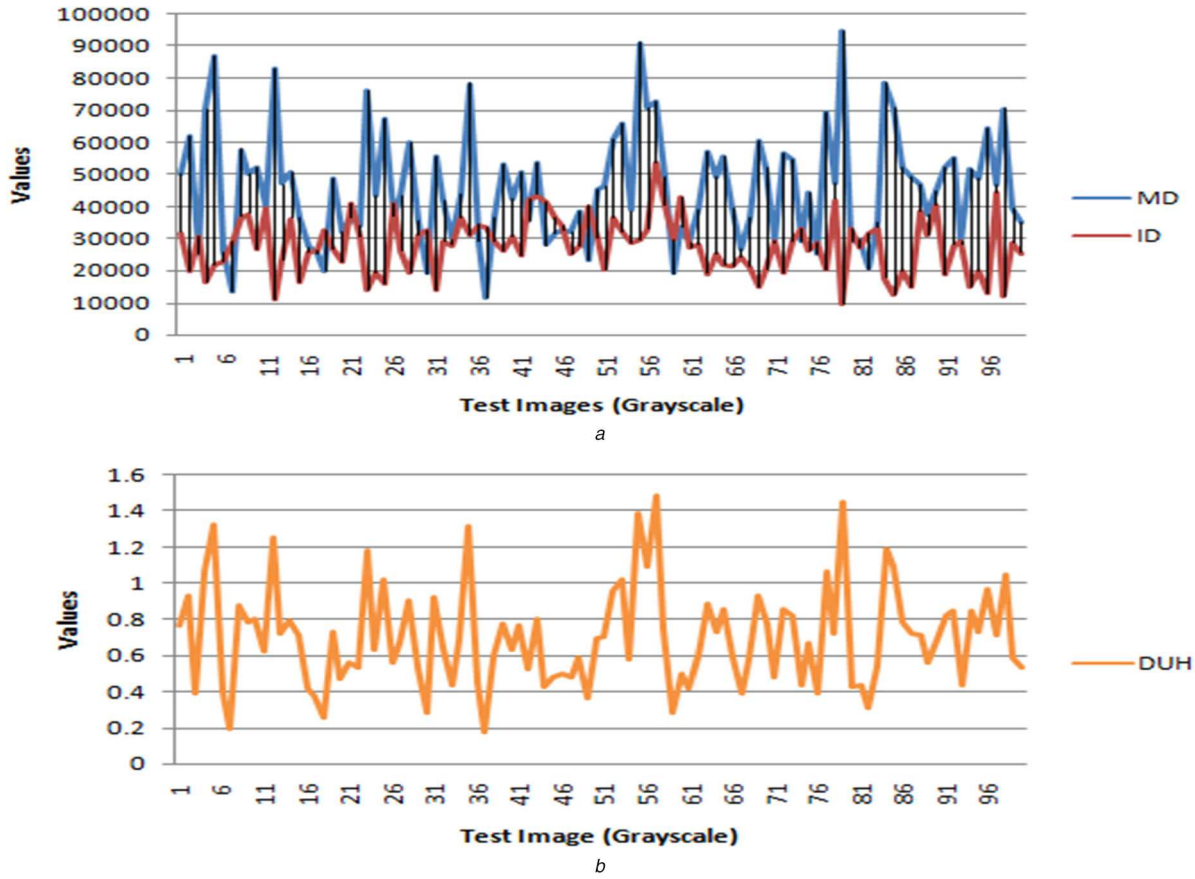


Fig. 16 *Encryption quality analysis*
(a) Encryption quality analysis: Maximum deviation and irregular deviation analysis, ***(b)*** Encryption quality analysis: Deviation from uniform histogram analysis

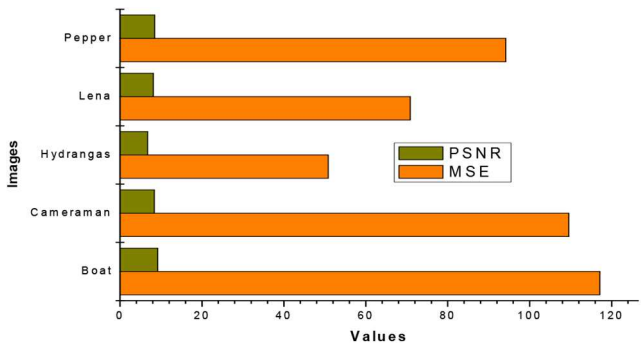


Fig. 17 *Error metric analysis – Greyscale images*

3.6 Occlusion & noise attack analyses

Attacks during the transmission of encrypted images result in the removal of certain sections or at the worst severe quality degradation of original images. Hence, it is necessary to evaluate the attack resistant capability of any cryptographic algorithm by applying various attacks such as occlusion (cropping) and noise with a threshold level [19]. To prove the attack resistant capability of the proposed work, occlusion and noise attacks have been performed on encrypted images. Figs. 20*a–d*, Fig. 21*a–c* and 22*a–c* depict the impact of cropping attack analysis with intentional removal of two different pixel sections on the greyscale and RGB encrypted images.

Noise attacks have been performed with three different noises namely Gaussian, Salt & Pepper and Speckle with various mean and variance values. Figs. 23*a–c*, Fig. 24*a–c*, 25*a–c* and 26*a–c* expresse the noise attack analysis on Lena image for the variance of 0.05. Though the attacks happened on encrypted images, the proposed algorithm can retrieve the original images with lesser amount of pixel degradation. From the results, it is verified that the proposed encryption scheme withstands against cropping and noise attacks.

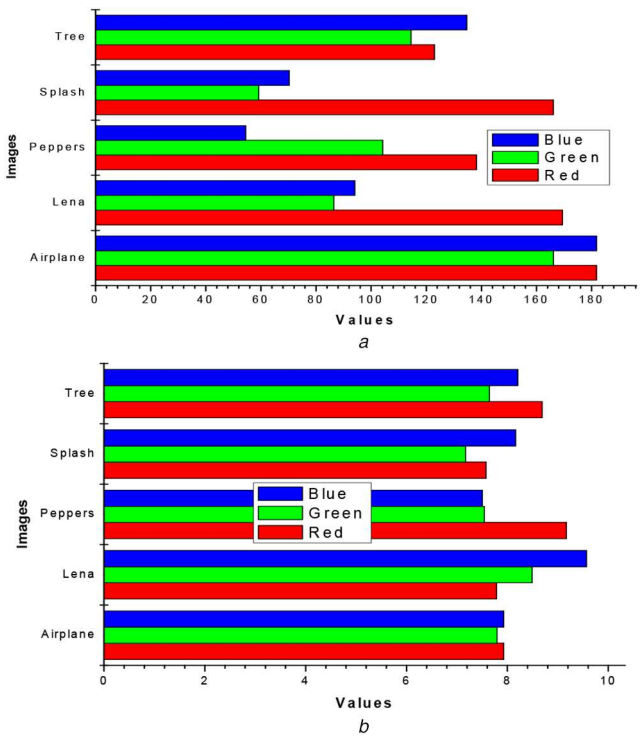


Fig. 18 *Error metric analysis – Colour images*
(a) MSE, ***(b)*** PSNR

3.7 NIST SP 800–22 batteries of test

National Institute of Standards and Technology (NIST) has released a test suite for evaluating the randomness of any RNG [23]. The ultimate goal of this evaluation is to determine the amount of randomness that can be generated from the source. It

© The Institution of Engineering and Technology 2020

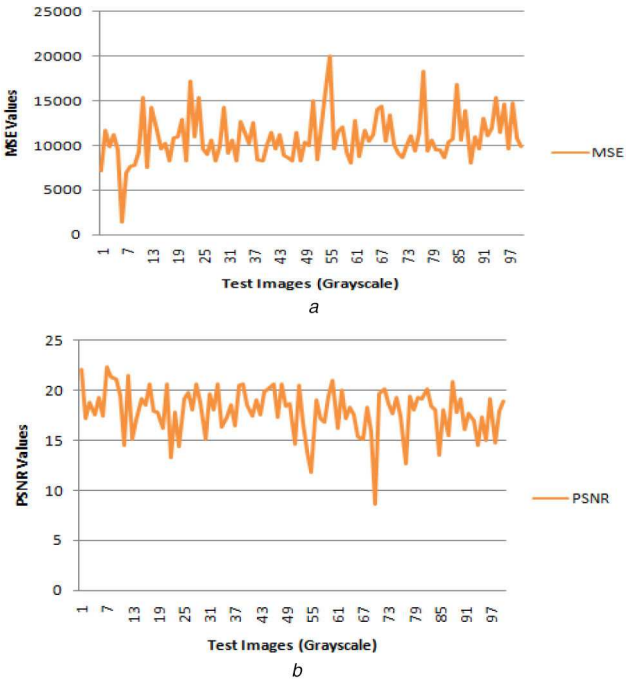


Fig. 19 *Error metric analysis for 100 greyscale images –*
(a) MSE, ***(b)*** PSNR

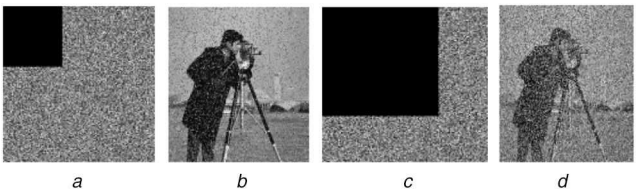


Fig. 20 *Cropping attack – Greyscale image*
(a) Level 1 cropping, ***(b)*** Decrypted image, ***(c)*** Level 2 cropping, ***(d)*** Decrypted image

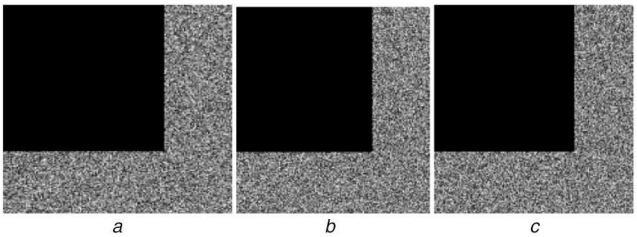


Fig. 21 *Cropping attack – Color image*
(a) Level 2 cropping on red plane, ***(b)*** Level 2 cropping on green plane, ***(c)*** Level 2 cropping on blue plane

analyses the uncertainty of generated random sequences. This test methodology can be used for analysing the cipher images wherein the amount of randomness among the pixels has been investigated. NIST has a few parameter settings to conduct the test which are given as follows:

Block frequency Test – Block length (M) = 128
Non-overlapping Template – Block length (m) = 9
Overlapping Template – Block length (m) = 9
Approximate Entropy Test – Block length (m) = 8
Serial Test – Block length (m) = 10
Linear Complexity Test – Block length (M) = 500

'To conduct this test, 65,536 × 8 bits and 65,536 × 3 × 8 bits were taken for greyscale and RGB colour images, respectively.

NIST SP 800-22 confirms the randomness when the P (Probability) value is greater than 0.01. Table 8 presents the test results of greyscale in which P value is in the acceptable range. Fig. 27 shows the range of P – values for each plane of RGB images with respect to list of available tests in NIST SP 800–22.

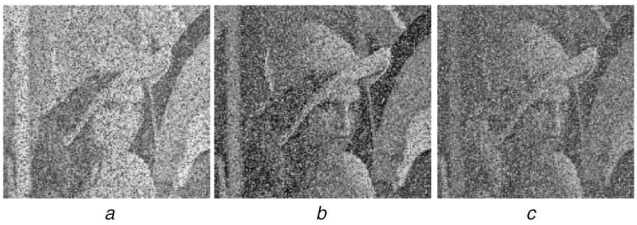


Fig. 22 *Cropping attack – Colour image*
(a) Decrypted red plane, ***(b)*** Decrypted green plane, ***(c)*** Decrypted blue plane

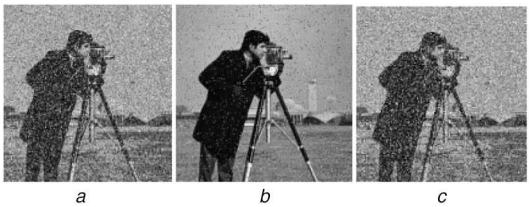


Fig. 23 *Noise attack on a test Grayscale image*
(a) Gaussian noise, ***(b)*** Salt & Pepper noise, ***(c)*** Speckle noise

Fig. 24 *Noise attack on a test Color image*
(a) Gaussian noise on red plane, ***(b)*** Salt & Pepper noise on red plane, ***(c)*** Speckle noise on red plane

Fig. 25 *Noise attack on Color image*
(a) Gaussian noise on green plane, ***(b)*** Salt & Pepper noise on green plane, ***(c)*** Speckle noise on green plane

Fig. 26 *Noise attack on Color image*
(a) Gaussian noise on blue plane, ***(b)*** Salt & Pepper noise on blue plane, ***(c)*** Speckle noise on blue plane

From the results, it can be understood that the proposed encryption scheme achieves good encryption.

3.8 Chosen-plaintext attack analysis

Since the XOR operation has been incorporated in the diffusion process, the effect of XOR operation on original and encrypted images requires to be analysed. From the perspective of cryptanalysts, chosen-plaintext attack can break the algorithm with the original image alone. Hence, that analysis has been conducted in which the original images of Cameraman and Peppers (I_1, I_2) were XORed and their corresponding encrypted images (C_1, C_2) were XORed. If both XORed images are equal, the algorithm is vulnerable to chosen-plaintext attack.

© The Institution of Engineering and Technology 2020

Table 8 NIST SP 800-22 results for greyscale images

Test parameters	*P* – values for 5 Test Greyscale images				
	Boat	Cameraman	Hydrangeas	Lena	Peppers
frequency	0.73991	0.350485	0.739918	0.35048	0.91141
block frequency	0.12232	0.739918	0.350485	0.35048	0.35048
cumulative sums – I	0.03517	0.350485	0.534146	0.73991	0.91413
cumulative sums – II	0.53414	0.534146	0.534146	0.21330	0.73991
runs	0.73991	0.739918	0.534146	0.53414	0.35048
longest run	0.12232	0.73991	0.12232	0.35048	0.35048
rank	0.01143	0.12232	0.35048	0.73991	0.91141
FFT	0.12232	0.73991	0.12232	0.73991	0.53414
non-overlapping template	0.91141	0.91141	0.91141	0.73991	0.91413
overlapping template	0.00204	0.06688	0.73991	0.21330	0.53414
approximate entropy	0.53414	0.53414	0.21330	0.35048	0.73991
serial – I	0.35048	0.73991	0.35048	0.73991	0.12232
serial – II	0.91141	0.35048	0.35048	0.91141	0.35048
linear complexity	0.53414	0.73991	0.53414	0.91141	0.35048

Figs. 28*a* and *b* confirm that the XORed results of ciphertext and plain text are not equal which means the algorithm is not vulnerable to the chosen-plaintext attack.

When compared to the earlier works of encryption reported in the introduction section, this proposed encryption scheme is well suited for greyscale and RGB images. The results evidence the superiority of the proposed methodology in terms of entropy, correction, differential, error metric and occlusion attack analyses. In addition, NIST SP 800–22 tests have been carried out to ensure the randomness of the encrypted images. The contributions of the proposed work are as follows:

i. Very first work on complete TRNG driven image encryption for greyscale and RGB colour images incorporating both confusion and diffusion whereas some of the earlier works used the TRNG only for diffusion [17–20].
ii. Simple TRNG architecture which consumes only 520 combinational logics and 524 logic registers on Cyclone IV EP4CE115F29C7 FPGA.
iii. Identical and Non-identical Ring Oscillator structures have been utilised for yielding better confusion and diffusion.
iv. The scheme offers large keyspace of $2^{(8\times65536+1)}$ which promises resistance against brute force attack.
v. This scheme yields a high entropy of 7.99 and near-zero correlation which confirms the resistance towards the statistical attacks when applied on more than 100 test images.
vi. The proposed non-chaotic approach improves the efficiency of the encryption and it makes the design free from intensive floating-point conversions.

4 Conclusion

A complete TRNG dependent image encryption has been suggested in this work. RO TRNGs using identical and non-identical rings of inverters played a vital role for confusion as well as diffusion operations. TRNGs were implemented on Altera Cyclone IV FPGA for the subsequent generation of synthetic images. Further, the proposed image encryption approach was tested on 100 greyscale and five RGB images to validate the strength through standard metrics. Though the proposed scheme offers good encryption by adopting a chaotic map free approach, it has some limitations due to the irreproducible property of TRNGs. Since the proposed work follows symmetric key encryption, it is necessary to share the keys between encryptor and decryptor securely. The synthetic image is larger almost equal to the size of the original image, and hence a secure key sharing mechanism must be developed. There is also a necessity to identify a proper compression algorithm for random synthetic images as they did not possess more number of redundant pixels which was evident from the entropy computation. The future direction promises the enhancement of proposed work by addressing certain limitations considering the growing hacking attempts on multimedia data.

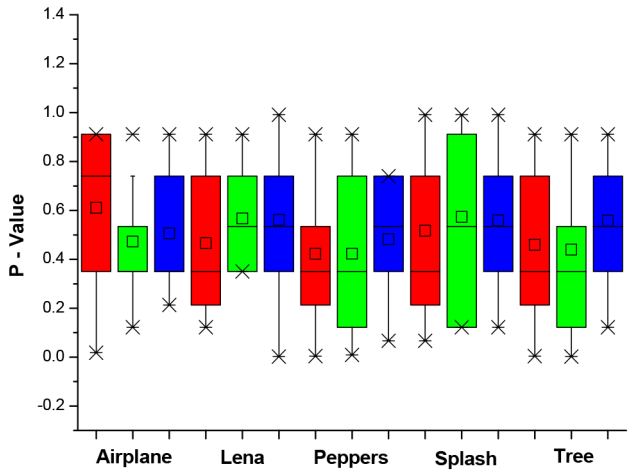


Fig. 27 *NIST SP 800–22 results for RGB images*

a *b*

Fig. 28 *Chosen plaintext attack*
(a) I_1 XOR I_2, ***(b)*** C_1 XOR C_2

5 Acknowledgment

The authors thank the Department of Science & Technology, New Delhi for the FIST funding (**SR/FST/ET-II/2018/221**). Also, the authors wish to thank the Intrusion Detection Lab at School of Electrical & Electronics Engineering, SASTRA Deemed University for providing infrastructural support to carry out this research work. The authors also acknowledge Mr. N. S. Srinath, Research Scholar, School of EEE, SASTRA Deemed to be University for his technical support.

6 References

[1] Hayat, U., Azam, N.A.: 'A novel image encryption scheme based on an elliptic curve', *Signal Process.*, 2019, **155**, pp. 391–402
[2] Xu, L., Li, Z., Li, J., *et al.*: 'A novel bit-level image encryption algorithm based on chaotic maps', *Opt. Lasers Eng.*, 2016, **78**, pp. 17–25
[3] Li, R., Liu, Q., Liu, L.: 'Novel image encryption algorithm based on improved logistic map', *IET Image Process.*., 2019, **13**, (1), pp. 125–134

© The Institution of Engineering and Technology 2020

[4] Askar, S.S., Karawia, A.A., Alammar, F.S.: 'Cryptographic algorithm based on pixel shuffling and dynamical chaotic economic map', *IET Image Process.*., 2018, **12**, (1), pp. 158–167
[5] Xu, L., Gou, X., Li, Z., *et al.*: 'A novel chaotic image encryption algorithm using block scrambling and dynamic index based di ff usion', *Opt. Lasers Eng. J.*, 2017, **91**, (November 2016), pp. 41–52
[6] Wu, J., Liao, X., Yang, B.: 'Colour image encryption based on chaotic systems and elliptic curve ElGamal scheme', *Signal Process.*, 2017, **141**, pp. 109–124
[7] Wang, X., Wang, S., Wei, N., *et al.*: 'A novel chaotic image encryption scheme based on hash function and cyclic shift', *IETE Tech. Rev. (Institution Electron. Telecommun. Eng. India)*, 2018, **4602**, (May), pp. 1–10
[8] Ur, A., Jan, R., Khan, S., *et al.*: 'A new image encryption scheme based on dynamic S-boxes and chaotic maps'. 3D Res., 77, 2016. Available at https://doi.org/10.1007/s13319-016-0084-9
[9] Belazi, A., El-Latif, A.A.A., Belghith, S.: 'A novel image encryption scheme based on substitution-permutation network and chaos,' *Signal Process.*, 2016, **128**, (Supplement C), pp. 155–170
[10] Kar, M., Mandal, M.K., Nandi, D., *et al.*: 'Bit-plane encrypted image cryptosystem using chaotic, quadratic, and cubic maps', *IETE Tech. Rev. (Institution Electron. Telecommun. Eng. India)*, 2016, **33**, (6), pp. 651–661
[11] Zhang, Y.: 'The image encryption algorithm with plaintext-related shuffling', *IETE Tech. Rev.*, 2016, **33**, (3), pp. 310–322
[12] Zhang, Y., Xiao, D.: 'An image encryption scheme based on rotation matrix bit-level permutation and block diffusion', *Commun. Nonlinear Sci. Numer. Simul.*, 2014, **19**, (1), pp. 74–82
[13] Preishuber, M., Hütter, T., Katzenbeisser, S., *et al.*: 'Depreciating motivation and empirical security analysis of chaos-based image and video encryption', *IEEE Trans. Inf. Forensics Secur.*, 2018, **13**, (9), pp. 2137–2150
[14] Norouzi, B., Mirzakuchaki, S.: 'Optik breaking an image encryption algorithm based on the new substitution stage with chaotic functions', *Opt. – Int. J. Light Electron Opt.*, 2016, **127**, (14), pp. 5695–5701
[15] Dou, Y., Liu, X., Fan, H., *et al.*: 'Cryptanalysis of a DNA and chaos based image encryption algorithm', *Opt. – Int. J. Light Electron Opt.*, 2017, **145**, pp. 456–464
[16] Zhang, Q., Guo, L., Wei, X.: 'A novel image fusion encryption algorithm based on DNA sequence operation and hyper-chaotic system', *Opt. – Int. J. Light Electron Opt.*, 2013, **124**, (18), pp. 3596–3600
[17] Liu, H., Kadir, A., Sun, X.: 'Chaos-based fast colour image encryption scheme with true random number keys from environmental noise', *IET Image Process.*., 2017, **11**, (55), pp. 324–332
[18] Rajagopalan, S., Rethinam, S., Arumugham, S., *et al.*: 'Networked hardware assisted key image and chaotic attractors for secure RGB image communication', *Multimedia Tools Appl.*, 2018, **77**, pp. 23449–23482
[19] Sundararaman, R., Sivaraman, R., Upadhyay, H.N., *et al.*: 'ON-chip peripherals are ON for chaos – an image fused encryption', *Microprocess. Microsyst.*, 2018, **61**, pp. 257–278
[20] Ravichandran, D., Rajagopalan, S., Upadhyay, H.N., *et al.*: 'Encrypted biography of biomedical image – a pentalayer cryptosystem on FPGA', *J. Signal Process. Syst.*, 2018, **91**, pp. 475–501
[21] Sunar, B., Martin, W.J., Stinson, D.R.: 'A provably secure true random number generator with built-in tolerance to active attacks', *IEEE Trans. Comput.*, 2007, **56**, (1), pp. 109–119
[22] Arumugham, S., Rajagopalan, S., Amirtharajan, R., *et al.*: 'Tamper-resistant secure medical image carrier: an IWT – SVD – chaos – FPGA combination', *Arab. J. Sci. Eng.*, 2019, **44**, (11), pp. 9561–9580
[23] Bassham, L.E., Rukhin, A.L., Soto, J., *et al.*: 'A statistical test suite for random and pseudorandom number generators for cryptographic applications'. Natl. Inst. Stand. Technol., Tech. Rep., Gaithersburg, MD, USA, Gaithersburg, MD, USA, 2010, April

© The Institution of Engineering and Technology 2020